S.C.O.U.T. Electrical · Progress Report

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Wed Oct 21 → Wed Oct 28, 2026

4 boards + 2 sensors
M0, Adalogger, ESP32-S3, LoRa wing · temperature + leak sensors ready to connect

Pending
second hardware order

Test plan
step-by-step bench tests ready

DONE THIS WEEK

Hardware

- ✓ Feather M0 + Adalogger: headers soldered, stacked (buoy main board)
- ✓ **Temperature probe:** leads soldered to jumpers, sealed with heat-shrink. **Ready to connect**
- ✓ **Leak sensor (SHT40):** header pins soldered. **Ready to connect**
- ✓ Shore station: ESP32-S3 + LoRa FeatherWing soldered

Design

- ✓ **Battery protection:** chose diodes (2 per D cell) so the two cells can never charge each other, as the battery maker requires for safety
- ✓ **Wiring schematic** (every part, pin and wire on one sheet): updated twice, now Draft C
- ✓ **Battery life** re-calculated for the new D cells: ≈ 28 months (worst case ≈ 15)
- ✓ **Leak sensor test plan:** one drop of water saturates a small box, so the test needs a large sealed one

Planning & documents

- ✓ **Electrical block diagram** (all parts and wires): reviewed by Dr. Krishnan, not yet approved
- ✓ **Bench test plan:** a step-by-step test for each part, plus 7 test programs and a tool to set the clock from a Windows laptop
- ✓ **Simulator** that predicts battery life, voltage while the radio transmits, and radio airtime
- ✓ **Thesis section** comparing options for the battery, its protection and the turbidity sensor

Parts & project

- ✓ Second hardware order pending: turbidity sensors, switches, coin cell
- ✓ **Shore station:** confirmed we already have every part, so no extra board to buy
- ✓ **GitHub:** block diagram and updated purchase log added
- ✓ **Task tracker (Linear):** closed 3 outdated tasks, updated 14

OBSTACLES

Lab access: weekend access issue, now resolved. I have 24/7 access going forward.

Shipping: second hardware order delayed in transit, due end of this week or next. Testing with parts in hand until then.

NEXT WEEK · OCT 28 → NOV 4

- ☐ **Thu** Check for short circuits with the multimeter, then power up the buoy board for the first time
- ☐ **Thu** First readings on the laptop: water temperature + humidity
- ☐ Solder antennas (buoy + shore) and the 3 wires that connect the shore station's radio to its board
- ☐ **First radio packet: buoy → shore station**
- ☐ Measure how much current the buoy draws (sets battery life), using the method Dr. Krishnan recommends
- ☐ When the second order arrives: coin cell + SD card in, then the turbidity sensor and its on/off switch

LAB SUPPLIES NEEDED

Wiring

- Jumper wires: male–male + male–female
- 1 extra breadboard (buoy and shore separate)
- Thin solid hookup wire (22–26 AWG) for the shore radio wires
- Resistors: 4.7 kΩ (temperature), 100 kΩ (switch)

Tools

- Heat gun from Jes (heat-shrink) + spare heat-shrink
- Soldering iron, solder, flux, solder wick
- Wire strippers + flush cutters
- Multimeter with µA/mA ranges, clip leads

Testing

- Bench power supply (3.6 V, for current tests)
- Data USB cables: micro-USB (M0), USB-C (shore)
- Cup of ice water + warm water (check the probe)
- Tape + marker to label wires